

HARIS WAHEED BHATTI

Nationality: Pakistani | Date of birth: 25/12/2001

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Work Authorisation: Pakistani national residing in Slovenia on a temporary student residence permit with full-time work rights; eligible for a post-study job-search residence permit from September 2026.

PROFILE

Haris Waheed Bhatti is a BIM and digital construction specialist completing a European Master's in Building Information Modelling at the University of Ljubljana, building on a Bachelor's in Civil Engineering from NUST, Pakistan. His work spans IFC-to-graph-database pipelines, LLM-based retrieval systems, and blockchain-based construction document management. He is seeking roles as a **BIM Software Developer**, **BIM Manager**, **Information Manager**, or **Digital Construction Specialist** in Slovenia.

EDUCATION

European Master in Building Information Modelling 15/11/2025 – 30/09/2026
University of Ljubljana *Slovenia*

- Field of study: Information Management in Construction
- Thesis:** Designing an IFC-to-Neo4j knowledge graph with an **LLM-based Retrieval-Augmented Generation (RAG)** pipeline for BIM data retrieval, implemented with **Claude** for AI-assisted development
- Modeling IFC building data as a property graph in Neo4j, preserving the hierarchical and relational structure of BIM entities
- Building an automated ingestion pipeline for bidirectional data flow between IFC files and the graph environment, enabling semantic querying and knowledge discovery across BIM datasets

BE Civil Engineering 09/2020 – 06/2024
National University of Science and Technology *Pakistan*

- Final grade: 3.25/4.00
- Thesis:** *Enhancing Construction Transparency and Dispute Resolution Through Blockchain-Based BIM Models* – developed a **Solidity** smart contract to digitally sign and manage construction documents, with off-chain storage linked to the blockchain for immutability and transparency
- Funded by a **PKR 100,000 grant** from the Pakistan Engineering Council; designed to streamline dispute resolution for construction stakeholders
- Link: bit.ly/49dUTmK | Contract: sepolia.arbiscan.io/address/0x8535...c867

PROJECTS

Pakistani-Themed Expo BIM Management | *IFC, Autodesk Construction Cloud* 28/10/2024 – 19/11/2024

- Spearheaded a BIM management project for a Pakistani-themed expo in Guimarães, Portugal, leveraging **IFC** and **Autodesk Construction Cloud** to develop a comprehensive technical design
- Facilitated collaboration between Architecture and MEP disciplines, creating a **federated model** that integrated diverse design inputs
- Conducted **clash detection** and **compliance checks** to ensure the model met all information requirements and design standards

EXPERIENCE

Contract Management Intern 07/08/2023 – 01/09/2023
Engineering and Management Services *Pakistan*

- Reviewed construction contract reports and arbitration case documentation under **FIDIC** conditions, and presented findings on FIDIC and arbitration procedures to the internal legal team

Research Intern (Part-time, Remote – Concurrent with Above) 17/07/2023 – 24/08/2023
Southern Illinois University Edwardsville *USA*

- Extracted and synthesized findings from academic articles and abstracts to support a faculty-mentored research proposal, applying feedback to refine presentation and analysis technique
- Supervisor: [linkedin.com/in/saadullah1](https://www.linkedin.com/in/saadullah1)

Civil Engineering Intern 11/07/2023 – 04/08/2023
Fatima Fertilizer Company *Pakistan*

- Modeled a 42m x 162m (6,804 m²) rock phosphate storage shed independently in 3D from 2D drawings, coordinating with stakeholders to refine the design

SKILLS

BIM & AEC Tools: IFC, Revit, Dynamo, Blender, Autodesk Construction Cloud, Three.js, WebIFC
Programming & Data: TypeScript, Python, Solidity, Neo4j, Knowledge Graphs
AI / LLM: Claude, Retrieval-Augmented Generation (RAG), Artificial Intelligence
Languages: Urdu (Native), English (Fluent)

LEADERSHIP & ACTIVITIES

Leadership & Extracurricular Activities	2020 – 2024
<i>National University of Science and Technology</i>	<i>Pakistan</i>
• Director, School Orientation Committee: Coordinated logistics and onboarding programming for incoming student cohorts during NUST Orientation • President, NUST 3v3 Madness Basketball Tournament: Organized team registration and event logistics for a university-wide competition	

CERTIFICATIONS

Coursera Certificates

Supervised Machine Learning: Regression and Classification	19/02/2023
• DeepLearning.AI course covering Python-based AI algorithms, applied in the "AI for Built Environment" course. (coursera.org/verify/T9ZJ8HYJX2E7)	
Management of Urban Infrastructures – Part 1	15/02/2024
• EPFL course on how cities operate and how infrastructure is built, operated, maintained, and governed. (coursera.org/verify/MWW4ZWUXSTDP)	
Innovative Governance of Large Urban Systems	21/02/2024
• Course on governing large urban cities while balancing infrastructure, policy, and citizen well-being. (coursera.org/verify/3VNHK3YRQ7MZ)	